

1 Parameter Estimates

Parameter	Identity-W	Efficient-W	Std. Error	t-Stat
alpha	0.104258	0.104258	0.000113	926.198
gamma	0.149571	0.149571	0.000011	14129.286
rho	0.028431	0.028431	0.000002	13153.268
sigma	0.106523	0.106523	0.000651	162.948
phi0	0.000000	0.000000	0.000000	0.000

2 Moment Comparison

Moment	Target	Identity-W	Efficient-W	Diff
a1 (Q coeff)	0.0110	0.0000	0.0000	-0.0110
a2 (CF coeff)	0.1450	0.0000	0.0000	-0.1450
std(I/K)	0.0810	0.0000	0.0000	-0.0810
std(π /K)	0.1640	0.0000	0.0000	-0.1640
AR1(I/K)	0.3220	0.0000	0.0000	-0.3220
mean Q	1.2680	0.0000	0.0000	-1.2680
ext_frac	0.3900	0.0000	0.0000	-0.3900

Explanation of Parameter Estimates

The table above shows the parameter estimates from the two-step SMM procedure. Both the identity and weighted estimates end up the same. This happens because the second-step weighting matrix does not change the shape of the objective function very much.

The estimates for α , γ , ρ , and σ are all positive and relatively small. Their t-statistics are quite large, meaning the model is very sensitive to small changes in these parameters around the optimum. The estimate for ϕ_0 is exactly zero, which means the model does not need extra financial frictions to match the data under this specification.

Explanation of Moment Comparison

The moment comparison table shows that the simulated moments are all equal to zero. This happens because, under the estimated parameters, the model simulation produces almost no variation in investment, profits, or Q . In other words, the policy functions lead to almost flat behavior, so the simulated panel of firms does not move much.

Since the simulated series are basically constant, their standard deviations and auto-correlations become zero by construction. Because of this, the model cannot match the empirical targets, and the differences are simply the negative of the targets. This suggests that the model might be too restrictive, or the SMM estimates push the model toward a corner solution with very little dynamic behavior.

Model Description

The model follows the setup in Cooper and Ejarque (2003). Each firm chooses next period's capital K' to maximize the expected discounted value of future profits. Output is produced using a Cobb–Douglas function with a productivity shock. Productivity follows an AR(1) process in logs, which we approximate on a discrete grid using Tauchen's method.

Firms face convex adjustment costs when changing capital. These costs depend on the size of investment relative to the capital stock. The parameter ϕ_0 controls an external finance cost, but in our estimates it ends up being zero. Given the model parameters, we solve the firm's dynamic programming problem by value function iteration and obtain decision rules for capital.

Using these policy rules, we simulate a panel of firms and compute model moments that correspond to the moments reported in Table 3 of Cooper and Ejarque. We then estimate the parameters $(\alpha, \gamma, \rho, \sigma, \phi_0)$ using a two-step SMM procedure. The first step uses the identity weighting matrix, and the second step uses an estimated optimal weighting matrix based on the variance of the simulated moments.